Lab 9 – Modeling Melanoma Rates

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Formatting Instructions



- Please submit your lab report as a **pdf** to Gradescope.
- When you upload to Gradescope, please **match pages** with the **question number**.
- Be sure that all **group members** are **added** in your submission to Gradescope (click view/edit group on the top right of the page once shown your final submission after matching pages).

Assignment Overview

- Melanoma rates in different U.S. states vary—but what might explain that variation?
- The *Melanoma.xlsx* file contains melanoma rates, in addition to other potentially related variables from various sources (most of which is from 2014-2017). Note this data includes the 48 contiguous states (excludes Alaska and Hawaii).
- We'll also be investigating a claim made in the cancercenter.com blog linked later regarding possible risk factors for melanomas. Does their claim stack up when we look at the data?

STEP 0

- **Pre-lab work**
  - Complete the pre-lab tutorials for Lab 9 first: https://stat212-learnr.stat.illinois.edu/
- **Download** the Melanoma.xlsx file to your computer and then **import** into your RStudio session.
- Create a new **R script** (or use the **RMarkdown file** if you are using that option)

Variables

Name: State Name

abb: State Abbreviation

mel: Number of new melanoma cases per 10,000 people each year

sun: The average sun exposure in each state, measured in kilojoules per square meter (kJ/m^2)

lat: Latitude of the center point of the state

lon: Longitude of the center point of the state

ocean: Whether the state borders an ocean or not

temp: Average daily high temperature year around

pop: State Population

density: Average number of state residents per square miles

age: Average age of state residents

white: Proportion of state residents that identify as both "white" and "non-hispanic"

income: Average income of state residents

(5pts) Question 1: Let's start by looking at the melanoma rates across U.S. states. By following these steps, we will create a map of the United States, and color each state by their melanoma rate. This should give us a little more insight to where melanoma rates are high.

Start by installing and librarying the package `"maps"`.

```
install.packages("maps")
library(maps)
```

This next step will call up some U.S. state map border data—let's save it as a dataframe called "MainStates." Feel free to click on it in your global environment to take a look. *Be sure you have libraried tidyverse first!*

```
MainStates = map_data("state")
```

Next, we want to merge this data frame into our Melanoma data frame. But to do that, we need a common column. Our `"Name"` column in Melanoma contains state names, but in MainStates, it's called `"region"`. Run the following command to change the MainStates data frame column to `"Name"`. Then, we will be able to easily match up these columns when we merge the data frames.

```
names(MainStates)[names(MainStates) == "region"] = "Name"
```

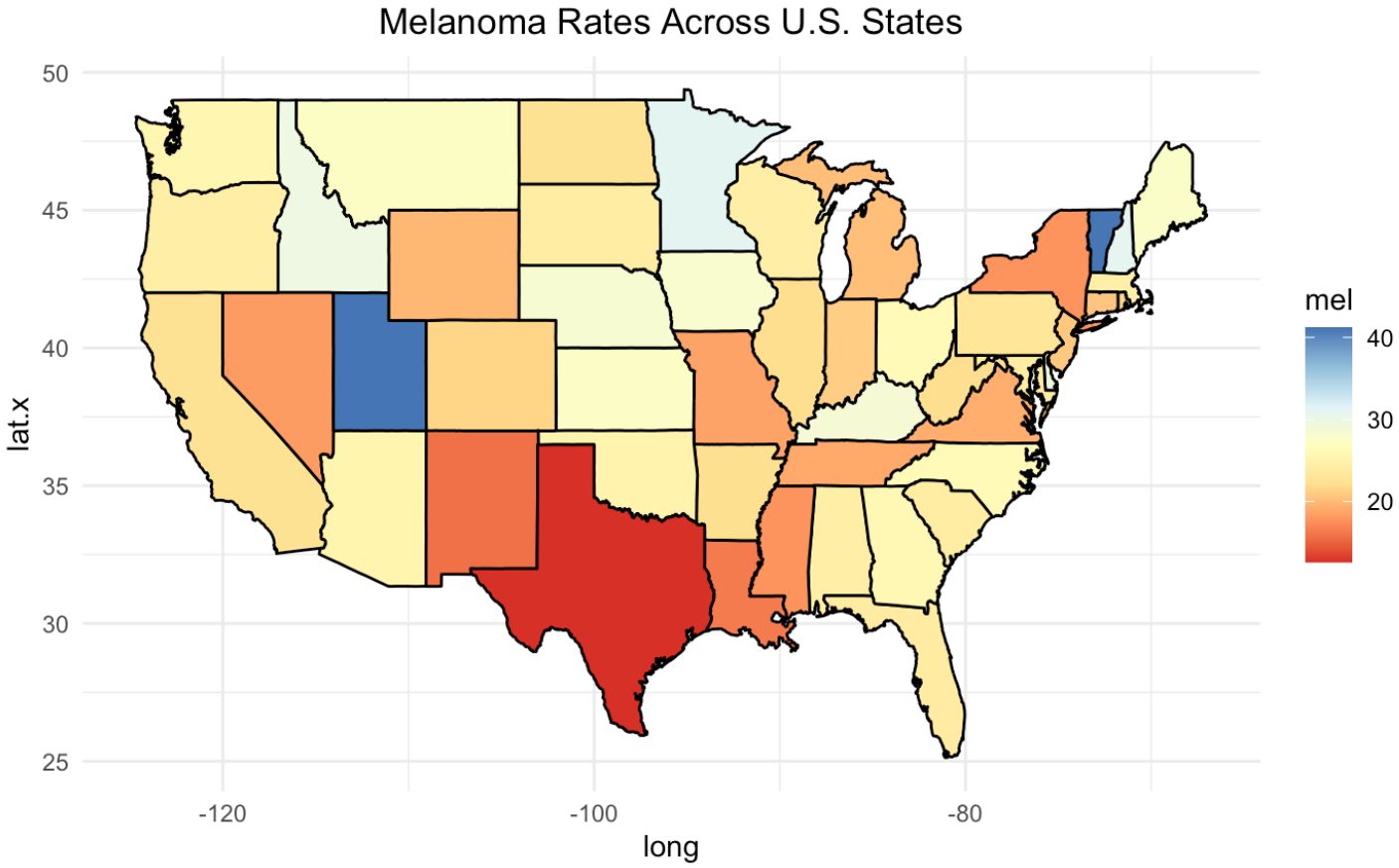
Run the following function to merge these two data frames. Go ahead and save this merged data frame under a different name (I suggest `Melanoma_Map`) so that we can still use the original data frame for other questions.

```
Melanoma_Map = inner_join(MainStates, Melanoma, by = "Name")
```

We're finally ready to create the map! Here are your instructions:

- Make a ggplot with **Melanoma_Map** as your data
- In the aes line, set x = long, y = lat.x, group = group, and fill = mel
    - Note, these are variable names from *Melanoma_Map*, not from Melanoma
- Use geom_polygon(), and set color = "black" inside this argument to insert black state borders
- Use a fill color palette (with distiller) to choose a custom color palette. The default palette is blues, but let's take this opportunity to pick a palette with more dynamic differences on each side of the scale.
- Add a background theme that provides a blank background (check the "Customizing ggplot2 tutorial, or search online for one!)
- Add an appropriate title

Insert your map graph




```r
```{r}
ggplot(data = Melanoma_Map, aes(x = long, y = lat.x, group = group, fill = mel)) +
  geom_polygon(color = "black") +
  scale_fill_distiller(palette = "RdYlBu", direction = 1) +
  theme_minimal() +
  ggtitle("Melanoma Rates Across U.S. States") +
  theme(plot.title = element_text(hjust = 0.5))

```
```

Which two states stand out as having the highest melanoma rates?

According to the states that appear blue in the graph, Vermont and Utah have the highest Melanoma rates.

Open the following short article about melanoma rates across U.S. States and read the first three paragraphs: https://www.cancercenter.com/community/blog/2016/05/unhealthiest-states-for-skin-cancer-risk-may-surprise-you

This article is considering different possible factors that might explain why some states have higher melanoma rates. One particular factor suggested by Dr. Charles Komen Brown is that **states with lower sun exposure on average tend to have higher melanoma rates.** He offers one possible *causal* explanation by saying *it is because residents in this state are not used to thinking about sun exposure and skin protection*.

Be sure to use the original Melanoma data you imported from this point on—the Melanoma_Map dataframe you created was only for Question 1.

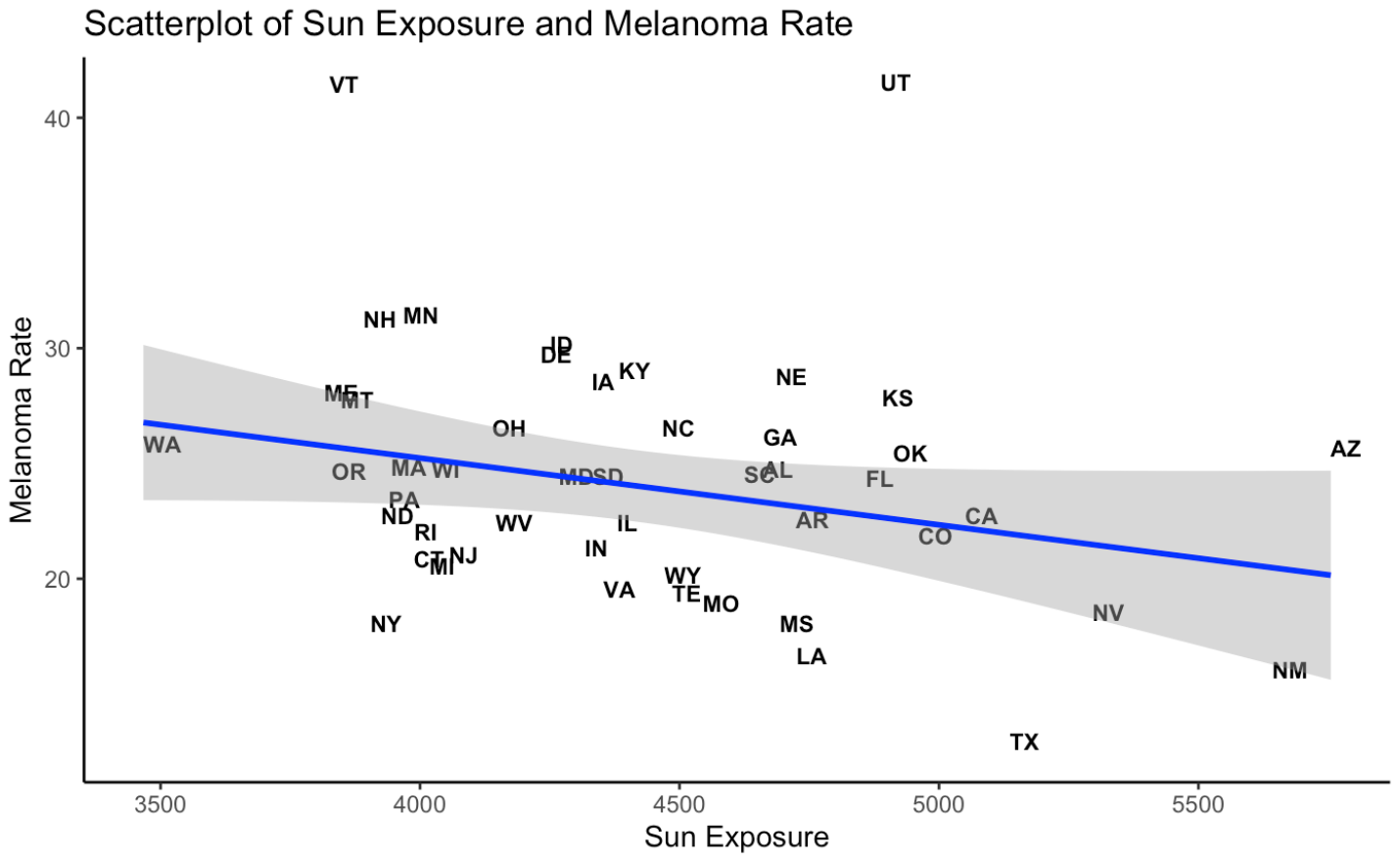
(5pts) Question 2: Let's look at the original Melanoma data with one row for each state. Create a scatterplot using sun exposure as the predictor variable and melanoma rate as the response variable. However, *in place* of using geom_point, substitute in the following geom code:

```
geom_text(aes(label=abb),hjust=0, vjust=0, size = 3, fontface = 2)
```

In addition:

- Add a best fit line (Standard error shading optional, Color choice up to you)
- Add a title
- Adjust the x and y axis titles to be fully written in 1-3 words (rather than the default variable abbreviations)
- Use the theme_classic() theme style (to provide a blank background)

Please include an image of this scatterplot in the report (code optional)



```r
ggplot(data = Melanoma, aes(x = sun, y = mel)) +
  geom_text(aes(label = abb), hjust = 0, vjust = 0, size = 3, fontface = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "blue") +
  theme_classic() +
  ggtitle("Scatterplot of Sun Exposure and Melanoma Rate") +
  xlab("Sun Exposure") +
  ylab("Melanoma Rate")
```

Which states have higher melanoma rates—those with more sun exposure on average, or less sun exposure?

According to the negative slope, states with less sun exposure are expected to have higher melanoma rates.

Which one or two states appear to have the largest residuals in this model?

Vermont (VT) and Utah (UT) have the largest residuals in this model.

(5pts) Question 3: Now, create a simple linear model with sun exposure as a predictor of melanoma rate.

Copy or screenshot the summary output of this model into your report *(starting with the row that says "Coefficient" through to the end).*

```
176 ```{r}
177 model <- lm(mel ~ sun, data = Melanoma)
178 summary(model)
179
180 ```
```

```
Call:
lm(formula = mel ~ sun, data = Melanoma)

Residuals:
    Min      1Q  Median      3Q     Max
-9.3419 -3.6914 -0.8312  2.4868 18.5344

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 36.813256   6.984338   5.271 3.52e-06 ***
sun         -0.002895   0.001574  -1.839   0.0723 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.334 on 46 degrees of freedom
Multiple R-squared:  0.0685,    Adjusted R-squared:  0.04825
F-statistic: 3.383 on 1 and 46 DF,  p-value: 0.07234
```

Interpret the multiple r-squared value *(Perhaps try the fill in the blank template we learned in chapter 13).*

In this linear regression model, the multiple R^2 value is 0.0685. This indicates that 6.85% of the variability in the response variable (melanoma rate) can be explained by the predictor variable (sun exposure) in the model. Therefore, the model provides a weak fit to the data, suggesting that 93.15% of the variability in melanoma rates is attributed to factors not included in the model.

Is there evidence that average state sun exposure and state melanoma rate have at least some linear association? *(Don't just say "reject" or "fail to reject" the null hypothesis—use the interpretational language we learned in Chapter 2: Little to no evidence, moderate evidence, strong evidence, very strong?).*

Based on the conventional significance level of 0.05, the p-value of 0.0723 is indeed greater than 0.05. Thus, there is little to no evidence to reject the null hypothesis. Specifically, this suggests that there is little to no evidence of a significant linear association between average state sun exposure and state melanoma rate based on the current model.

(5pts) Question 4: Let's look at some other possible predictors of melanoma rate: Average temperature, Average age, and Proportion of white, non-hispanic residents. Create a simple linear model for each of these numeric predictors (i.e., create three simple linear models) and look at the output.

```
Call:
lm(formula = mel ~ temp, data = Melanoma)

Residuals:
    Min      1Q  Median      3Q     Max
-8.7137 -3.9797 -0.3196  3.2586 16.3181

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  35.5865     5.3592   6.640 3.16e-08 ***
temp         -0.2203     0.1013  -2.175   0.0348 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.262 on 46 degrees of freedom
Multiple R-squared:  0.09328,   Adjusted R-squared:  0.07357
F-statistic: 4.732 on 1 and 46 DF,  p-value: 0.03478


Call:
lm(formula = mel ~ age, data = Melanoma)

Residuals:
    Min      1Q  Median      3Q     Max
-11.012  -3.570   0.047   2.338  18.012

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  19.7300    13.5648   1.455    0.153
age           0.1116     0.3500   0.319    0.751

Residual standard error: 5.52 on 46 degrees of freedom
Multiple R-squared:  0.002203,  Adjusted R-squared:  -0.01949
F-statistic: 0.1016 on 1 and 46 DF,  p-value: 0.7514


Call:
lm(formula = mel ~ white, data = Melanoma)

Residuals:
    Min      1Q  Median      3Q     Max
-7.3349 -2.9884 -0.2833  2.6083 15.4837

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   10.160      3.377   3.009 0.004242 **
white         19.867      4.733   4.198 0.000122 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.699 on 46 degrees of freedom
Multiple R-squared:  0.277,     Adjusted R-squared:  0.2613
F-statistic: 17.62 on 1 and 46 DF,  p-value: 0.000122
```

```r
model_temperature <- lm(mel ~ temp, data = Melanoma)
model_age <- lm(mel ~ age, data = Melanoma)
model_proportion_white <- lm(mel ~ white, data = Melanoma)

summary(model_temperature)
summary(model_age)
summary(model_proportion_white)
```

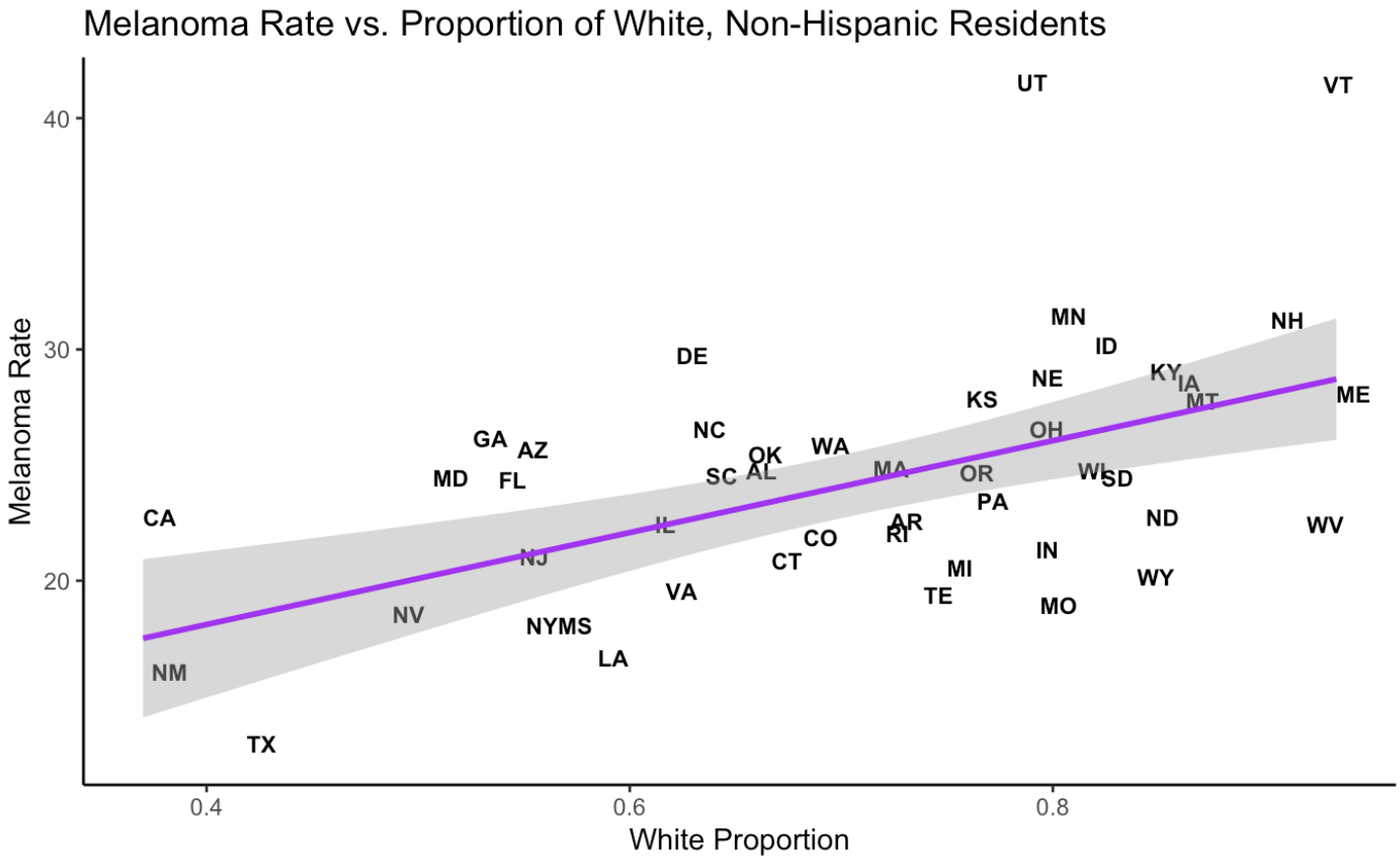
Are any of these three variables stronger predictors of melanoma rate than sun exposure? Explain how you made that determination.

Comparing the coefficient: Proportion of White, Non-Hispanic Residents (coefficient 19.867) has the largest magnitude among all variables. Average Temperature (coefficient -0.2203) has a moderate magnitude. Sun Exposure (coefficient −0.002895) has the smallest magnitude.

Comparing the p-values: Proportion of White, Non-Hispanic Residents (p-value 0.000122) has the smallest p-value, indicating strong statistical significance. Average Temperature (p-value 0.0348) has a moderately small p-value. Sun Exposure (p-value 0.0723) has a larger p-value, suggesting weaker statistical significance.

By the above comparison, Proportion of White, Non-Hispanic Residents appears to be a stronger predictor of melanoma rate than sun exposure, based on both the magnitude of the coefficient and the strong statistical significance. Besides, Average Temperature also shows a moderate strength in prediction. Sun exposure has the smallest magnitude of the coefficient and a larger p-value, indicating a comparatively weaker predictor.

Create a scatterplot of the strongest predictor of these three. Again, use state abbreviations in place of points. Follow all formatting guidelines from question 2. **Include an image of this plot in your report.**




```r
ggplot(Melanoma, aes(x = white, y = mel)) +
  geom_text(aes(label = abb), hjust = 0, vjust = 0, size = 3, fontface = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "purple") +
  labs(
    title = "Melanoma Rate vs. Proportion of White, Non-Hispanic Residents",
    x = "White Proportion",
    y = "Melanoma Rate"
  ) +
  theme_classic()
```

(5pts) Question 5: While it is possible to create a model with two numeric predictors, let's focus on the simpler case of using one numeric predictor with one binary predictor.

Using an `ifelse` statement, create a new variable in the dataset called `white_binary` that records "High" if the state's proportion of residents being "white, non-hispanic" is above 0.7 and "Low" if the proportion is below 0.7.

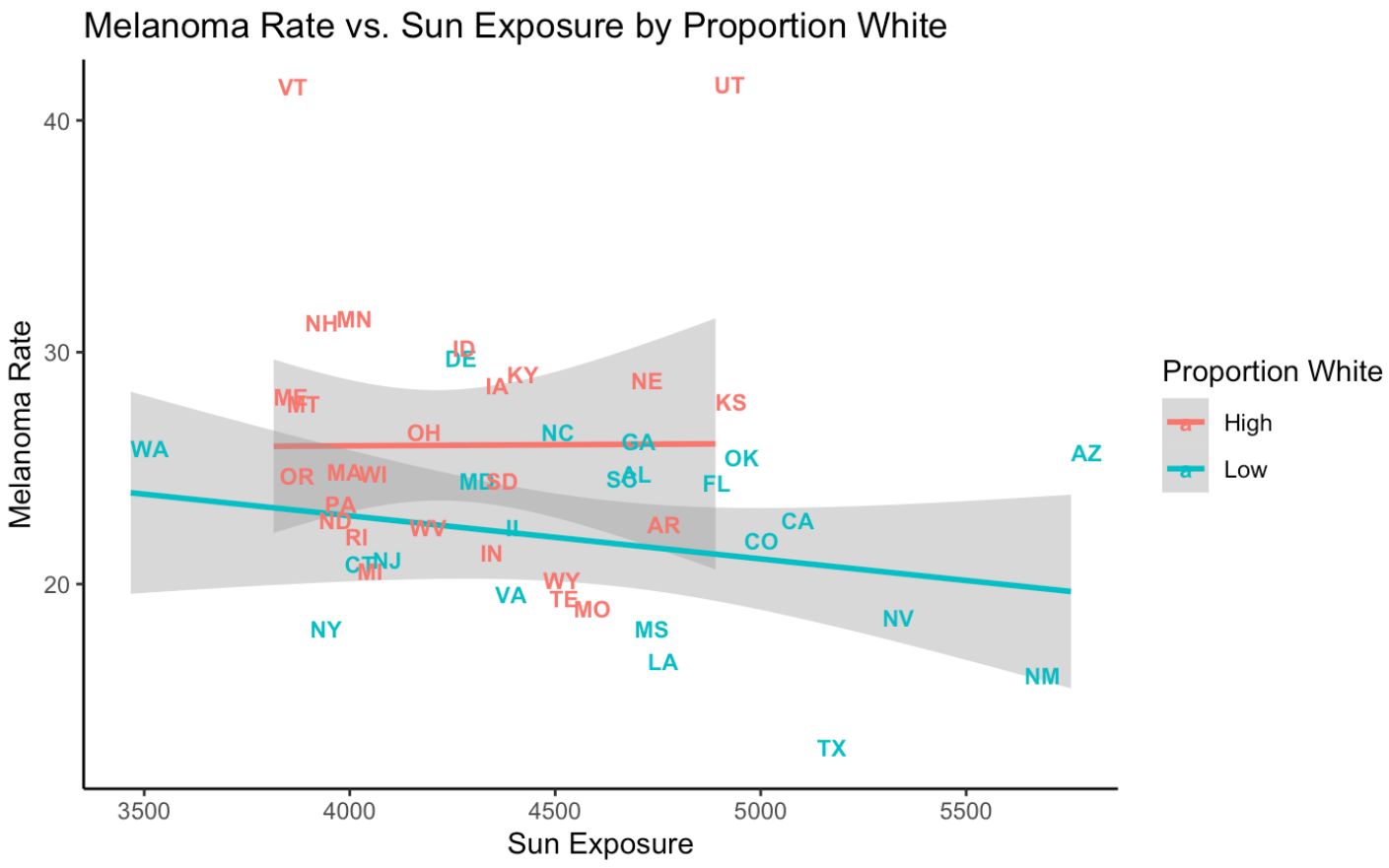
Include your ifelse statement code

```r
Melanoma$white_binary <- ifelse(Melanoma$white > 0.7, "High", "Low")
```

Create a scatterplot with Sun exposure on the x axis, color the data points based on whether the proportion of white residents is above 0.7 or not, and put Melanoma rate on the y axis.

- Continue to use the state abbreviations in place of points.
- Add best fit lines for each group (i.e., visualize what an **interaction** model would look like).
- Add an appropriate title
- Adjust the axes labels. Also adjust the color legend to say "Proportion White." You can do that by entering a color = "…" argument into the labs function.

Include an image of your plot here *and* your code for this plot.



```r
ggplot(Melanoma, aes(x = sun, y = mel, color = white_binary)) +
  geom_smooth(method = "lm", aes(group = white_binary)) +
  geom_text(aes(label = abb), hjust = 0, vjust = 0, size = 3, fontface = 2) +
  labs(
    title = "Melanoma Rate vs. Sun Exposure by Proportion White",
    x = "Sun Exposure",
    y = "Melanoma Rate",
    color = "Proportion White"
  ) +
  theme_classic()
```

(5pts) Question 6: In the previous question, we plotted an interaction model (allowing for different slopes). Now, create an linear model with these same 2 predictors (the sun exposure variable, and white_binary) that includes an interaction term.

Copy or screenshot the summary output of your **interaction model** into the report.

```
265 ```{r}
266 model_interaction <- lm(mel ~ sun * white_binary, data = Melanoma)
267 summary(model_interaction)
268 ```
```

```
Call:
lm(formula = mel ~ sun * white_binary, data = Melanoma)

Residuals:
    Min      1Q  Median      3Q     Max
-8.2302 -3.8440  0.0417  2.6598 15.1521

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)          2.559e+01  1.295e+01   1.976   0.0544 .
sun                  9.438e-05  3.062e-03   0.031   0.9755
white_binaryLow      4.804e+00  1.603e+01   0.300   0.7658
sun:white_binaryLow -1.956e-03  3.669e-03  -0.533   0.5968
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.155 on 44 degrees of freedom
Multiple R-squared:  0.1678,    Adjusted R-squared:  0.1111
F-statistic: 2.958 on 3 and 44 DF,  p-value: 0.04258
```

Does the relationship between a state’s average sun exposure and that state’s melanoma rate appear to be dependent on the proportion of white, non-hispanic residents? *Justify your answer using your model summary. It also may help you to consult your graph from the previous question to see how this visual relates to the model output.*

First, the interaction term “sun:white_binaryLow” has a coefficient of -0.001956 with a p-value of 0.5968. The p-value is greater than the conventional significance level of 0.05, indicating that the interaction term is not statistically significant.

Furthermore, because the best fit lines in the graph did not visibly intersect, this aligns with weak significance in the interaction term within the statistical model.

Thus, both our statistical model and the graphical representation suggest that there is no strong evidence to suggest a significant variation in the relationship between sun exposure and melanoma rate based on different proportions of white, non-Hispanic residents.

(5pts) Question 7: Now, create an additive model for Sun exposure and white_binary as predictors of Melanoma rate, and consult this model to answer the following questions:

Copy or screenshot the summary output of your **additive model** into the report

```r
```{r}
additive_model <- lm(mel ~ sun + white_binary, data = Melanoma)
summary(additive_model)
```
```

```
Call:
lm(formula = mel ~ sun + white_binary, data = Melanoma)

Residuals:
   Min     1Q Median     3Q    Max
-8.525 -3.958  0.067  2.735 16.066

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     31.326485   7.127412   4.395 6.68e-05 ***
sun             -0.001267   0.001674  -0.757   0.4530
white_binaryLow -3.691684   1.643171  -2.247   0.0296 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.113 on 45 degrees of freedom
Multiple R-squared:  0.1624,    Adjusted R-squared:  0.1252
F-statistic: 4.364 on 2 and 45 DF,  p-value: 0.01853
```

Is there evidence that sun exposure has a linear relationship with melanoma rate if we already know whether the proportion white, non-hispanic residents in a state is above or below 0.7? *Briefly justify*

The coefficient for sun is estimated to be -0.001267 with a p-value of 0.4530. The p-value is greater than 0.05, indicating that the coefficient is not statistically significant. Thus, there is no strong evidence to support a linear relationship between sun exposure and melanoma rate in the context of knowing whether the proportion of white, non-Hispanic residents in a state is above or below 0.7.

The coefficient for white_binaryLow is -3.691684 with a statistically significant p-value of 0.0296. This suggests that the relationship between melanoma rate and the proportion of white, non-Hispanic residents is significant, but it doesn't provide strong evidence for a linear relationship with sun exposure.

Contextually, how might the proportion of white, non-hispanic residents act as a *confounder* to the association between sun exposure and melanoma rate? *(Check back to Chapter 9 and look for the confounding diagram. Think about what that might look like here!).*

Sun exposure is considered the exposure variable X; Melanoma rate is the outcome variable Y; The proportion of white, non-Hispanic residents is denoted as W.

There is an association between W and Y, as indicated by the statistically significant coefficient for white_binaryLow in the model. Sun exposure X may influence the proportion of white, non-Hispanic residents W in a state.

If sun exposure (X) affects the proportion of white, non-Hispanic residents (W), and W is associated with melanoma rate (Y), then W acts as a confounder in the association between X and Y. In other words, the proportion of white, non-Hispanic residents is a third variable that influences both sun exposure and the melanoma rate.

Failure to account for the proportion of white, non-Hispanic residents as a confounder may lead to a biased estimate of the true association between sun exposure and melanoma rate. The observed association between sun exposure and melanoma rate may be confounded by the influence of the proportion of white, non-Hispanic residents.